Bash Scripting | Cronjobs

Open gitbash and connect to AWS ec2 server (ssh -I key.pem ec2-user@ip4-public)

```
HomePC@PC MINGW64 ~/Downloads
$ ssh -i bash.pem ec2-user@52.65.212.45

A newer release of "Amazon Linux" is available.
  Version 2023.12.20261014:
Run "/usr/bin/dnf check-release-update" for full release and version update info
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 16 06:57:12 2026 from 103.203.172.230
[ec2-user@ip-172-31-35-54 ~]$
```

Check your shell using a command (echo $0)

```
[root@ip-172-31-35-54 ec2-user]# echo $0
bash
[root@ip-172-31-35-54 ec2-user]#
```

Created a directory path to save all execution in one directory (mkdir -p viquar/bash/)

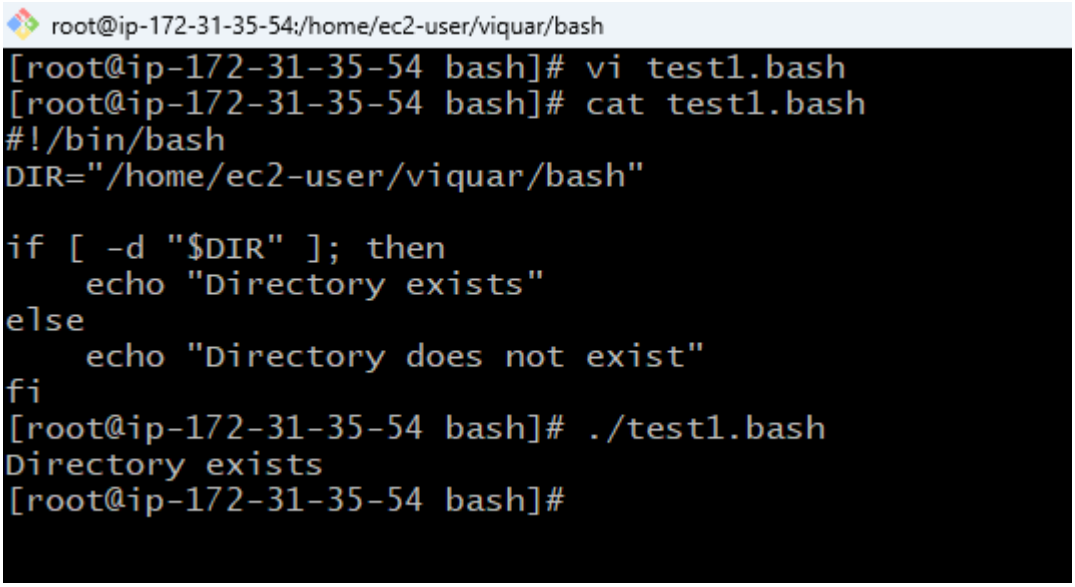
```
[root@ip-172-31-35-54 ec2-user]# mkdir -p viquar/bash/
[root@ip-172-31-35-54 ec2-user]# pwd
/home/ec2-user
[root@ip-172-31-35-54 ec2-user]# ls
viquar
[root@ip-172-31-35-54 ec2-user]# cd viquar/bash/
[root@ip-172-31-35-54 bash]# ls
[root@ip-172-31-35-54 bash]#
```

1. Create a bash script to **check if a directory is available or not**.
   Step 1: Created a file using a command vi (test1.bash) and wrote a script

```
#!/bin/bash
DIR="/home/ec2-user/viquar/bash"

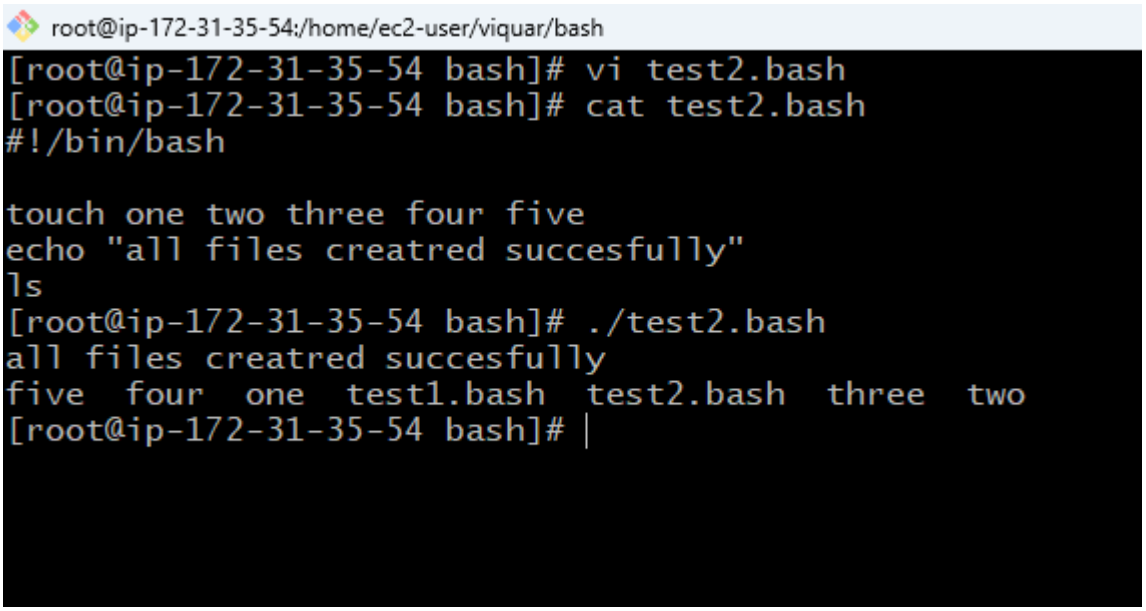
if [ -d "$DIR" ]; then
   echo "Directory exists”
else
   echo "Directory does not exist"
fi
```

Step 2: Change the permissions to file using a command (chmod 755 test1.bash)
Step 3: see the contain of a file using a command (cat test1.bash)
Step 4: Executed the script using a command (./test1.bash)
RESULTS BELOW



2. Create a bash script to **create multiple files**.
Step 1: Created a file using a command (vi test2.bash) and wrote a script
#!/bin/bash

touch one two three four five
echo "All files created successfully!"
ls
Step 2: changed permissions using a command (chmod 755 test2.bash)
Step 3: checked the content of file using a command (cat test2.bash)
Step 4: Executed the script using a command (./test2.bash)
RESULTS BELOW

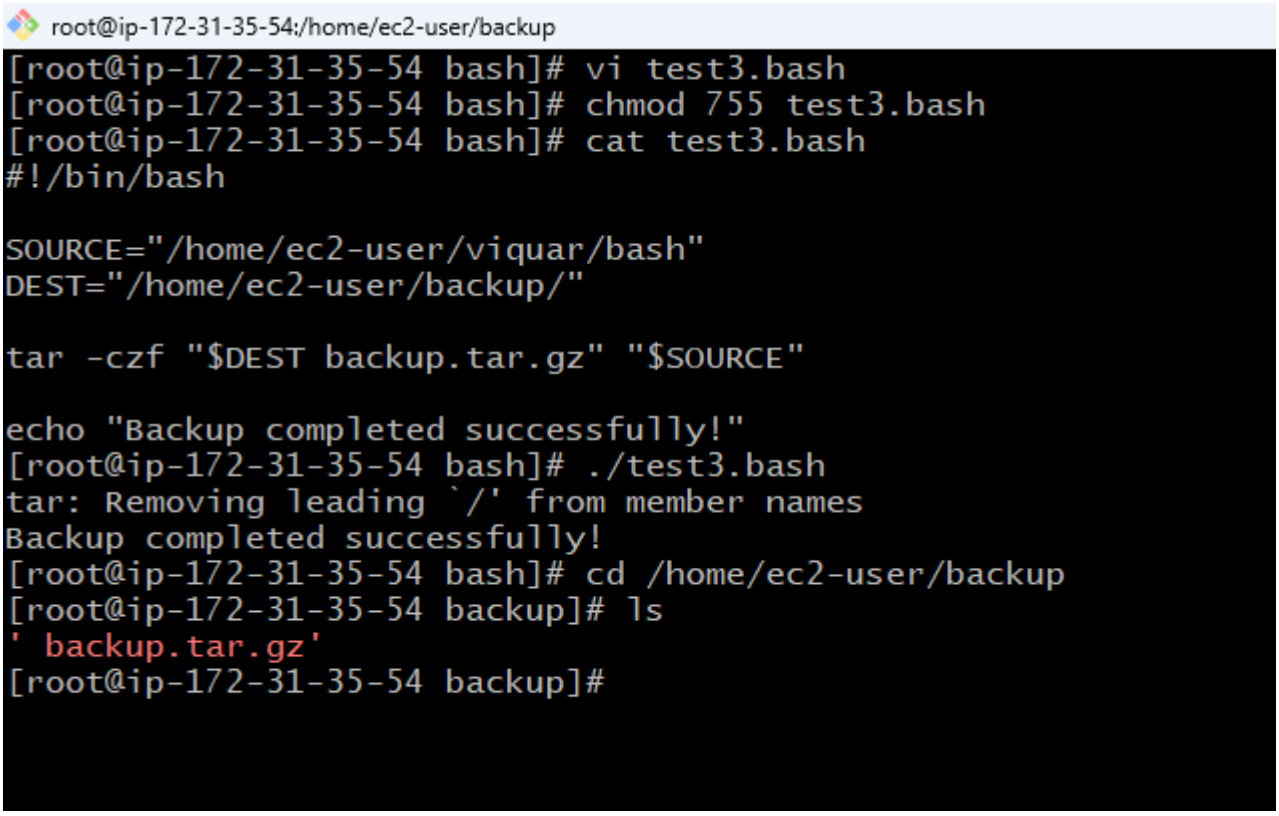


3. Create a bash script to **take a backup of a directory**.
   Step 1:
   #!/bin/bash

   SOURCE=”/home/ec2-user/viquar/nash”
   DEST=”/home/ec2-user/backup/”

   Tar -czf “$DEST backup.tar.gz” “SOURCE”

   echo "Backup completed successfully!"
   Step 2: Check the dir path using a command (cd home/ec2-user/backup)
   Step 3: used command ls to check file available in backup directory
   RESULTS BELOW



4. Create a bash script to **install Nginx on an EC2 server**.
   Step 1: Created a file using a command vi (test4.bash) and wrote a script
   #!/bin/bash

   sudo yum install nginx -y

   sudo systemctl start nginx

   sudo systemctl enable nginx

   echo "Nginx installed successfully!"

Step 2: Change the permissions to file using a command (chmod 755 test4.bash)

Step 3: see the contain of a file using a command (cat test4.bash)

Step 4: Executed the script using a command (./test4.bash)

Step 5: After succesfull installation I checked using a command (systemctl status nginx)

RESULTS BELOW

```
root@ip-172-31-35-54:/home/ec2-user/backup
[root@ip-172-31-35-54 backup]# vi test4.bash
[root@ip-172-31-35-54 backup]# chmod test4.bash
chmod: missing operand after ‘test4.bash’
Try 'chmod --help' for more information.
[root@ip-172-31-35-54 backup]# chmod 775 test4.bash
[root@ip-172-31-35-54 backup]# cat test4.bash
#!/bin/bash

yum install nginx -y

systemctl start nginx

systemctl enable nginx

echo "Nginx installed successfully!"
[root@ip-172-31-35-54 backup]# ./test4.bash
Last metadata expiration check: 1 day, 2:37:52 ago on Thu Oct 15 09:36:54 2026.
Dependencies resolved.
================================================================================
 Package                         Architecture          Version
================================================================================
Installing:
 nginx                           x86_64                1:1.30.5-1.amzn2023.0.1
Installing dependencies:
 generic-logos-httpd             noarch                18.0.0-12.amzn2023.0.3
 gperftools-libs                 x86_64                2.9.1-1.amzn2023.0.3
 libunwind                       x86_64                1.4.0-5.amzn2023.0.3
 nginx-core                      x86_64                1:1.30.5-1.amzn2023.0.1
 nginx-filesystem                noarch                1:1.30.5-1.amzn2023.0.1
 nginx-mimetypes                 noarch                2.1.49-3.amzn2023.0.3

Transaction Summary
================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
```

```
================================================================================

Installed:
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                gperftools-l
  libunwind-1.4.0-5.amzn2023.0.3.x86_64                            nginx-1:1.30
  nginx-core-1:1.30.5-1.amzn2023.0.1.x86_64                        nginx-filesy
  nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/
Nginx installed successfully!
[root@ip-172-31-35-54 backup]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disa
     Active: active (running) since Fri 2026-10-16 12:14:49 UTC; 1min 23s ago
   Main PID: 53244 (nginx)
      Tasks: 3 (limit: 1059)
     Memory: 3.4M
        CPU: 77ms
     CGroup: /system.slice/nginx.service
             ├─53244 "nginx: master process /usr/sbin/nginx"
             ├─53245 "nginx: worker process"
             └─53246 "nginx: worker process"

Oct 16 12:14:49 ip-172-31-35-54.ap-southeast-2.compute.internal systemd[1]: Start
Oct 16 12:14:49 ip-172-31-35-54.ap-southeast-2.compute.internal nginx[53150]: ngi
Oct 16 12:14:49 ip-172-31-35-54.ap-southeast-2.compute.internal nginx[53150]: ngi
Oct 16 12:14:49 ip-172-31-35-54.ap-southeast-2.compute.internal systemd[1]: Start

[root@ip-172-31-35-54 backup]#
```

5. Create a bash script to **install Apache Tomcat on an EC2 server**.

   Step 1: Created a file using a command vi (test5.bash) and wrote a script

   #!/bin/bash

   sudo yum install tomcat10 -y

   sudo systemctl start tomcat10
   sudo systemctl enable tomcat10

   echo "tomcat installed successfully!"

   Step 2: Change the permissions to file using a command (chmod 755 test5.bash)

   Step 3: see the contain of a file using a command (cat test5.bash)

   Step 4: Executed the script using a command (./test5.bash)

   Step 5: After succesfull installation I checked using a command (systemctl status tomcat10)

   RESULTS BELOW

```
root@ip-172-31-35-54:/home/ec2-user/viquar/bash
[root@ip-172-31-35-54 bash]# vi test5.bash
[root@ip-172-31-35-54 bash]# chmod 755 test5.bash
[root@ip-172-31-35-54 bash]# cat test5.bash
#!/bin/bash

sudo yum install tomcat10 -y

sudo systemctl start tomcat10
sudo systemctl enable tomcat10

echo "tomcat installed successfully!"
[root@ip-172-31-35-54 bash]# ./test5.bash
Last metadata expiration check: 1 day, 2:51:38 ago on Thu Oct 15 09:36:54 2
Dependencies resolved.
================================================================================
 Package                                 Architecture       Version
================================================================================
Installing:
 tomcat10                                noarch             1:10.1.60-
Installing dependencies:
 apr                                     x86_64             1.7.5-1.am
 ecj                                     noarch             1:4.23-1.a
 java-17-amazon-corretto-headless        x86_64             1:17.0.20+
 javapackages-tools                      noarch             6.0.0-7.am
 tomcat10-el-5.0-api                     noarch             1:10.1.60-
 tomcat10-jsp-3.1-api                    noarch             1:10.1.60-
 tomcat10-lib                            noarch             1:10.1.60-
 tomcat10-servlet-6.0-api                noarch             1:10.1.60-
Installing weak dependencies:
 tomcat-native                           x86_64             2.0.16-1.a

Transaction Summary
================================================================================
Install  10 Packages
```

```
tomcat installed successfully!
[root@ip-172-31-35-54 bash]# systemctl status tomcat10
● tomcat10.service - Apache Tomcat 10 Web Application Container
     Loaded: loaded (/usr/lib/systemd/system/tomcat10.service; enabled;
     Active: active (running) since Fri 2026-10-16 12:28:41 UTC; 1min 3
   Main PID: 76205 (java)
      Tasks: 29 (limit: 1059)
     Memory: 102.4M
        CPU: 2.700s
     CGroup: /system.slice/tomcat10.service
             └─76205 /usr/lib/jvm/jre/bin/java -Djavax.sql.DataSource.F

Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
Oct 16 12:28:42 ip-172-31-35-54.ap-southeast-2.compute.internal server[
lines 1-20/20 (END)
```

6. Create a bash script to **check if the Nginx service is running**, if not running then script should start the service.

   Step 1: Created a file using a command vi (test6.bash) and wrote a script

   ```
   #!/bin/bash

   if systemctl status nginx | grep -q "active (running)"
   then
     echo Nginx is already running
   else
     sudo systemctl start nginx
     echo "Nginx service started"
   fi
   ```

   Step 2: Change the permissions to file using a command (chmod 755 test6.bash)

   Step 3: see the contain of a file using a command (cat test6.bash)

   Step 4: Executed the script using a command (./test6.bash)

   Note: if your using conditional scripting make sure to end script with fi

   RESULTS BELOW

```
root@ip-172-31-35-54:/home/ec2-user/viquar/bash
[root@ip-172-31-35-54 bash]# vi test6.bash
[root@ip-172-31-35-54 bash]# chmod 755 test6.bash
[root@ip-172-31-35-54 bash]# cat test6.bash
#!/bin/bash

if systemctl status nginx | grep -q "active (running)"
then
    echo "Nginx is already running"
else
    sudo systemctl start nginx
    echo Nginx service started
fi
[root@ip-172-31-35-54 bash]# ./test6.bash
Nginx is already running
[root@ip-172-31-35-54 bash]#
```

7. Create a bash script for a **calculator**.

   Step 1: Created a file using a command vi (test7.bash) and wrote a script

   #!/bin/bash

   echo "Enter first number:"
   read a

   echo "Enter second number:"
   read b

   echo "Addition: $((a + b))"
   echo "Subtraction: $((a - b))"
   echo "Multiplication: $((a * b))"
   echo "Division: $((a / b))"

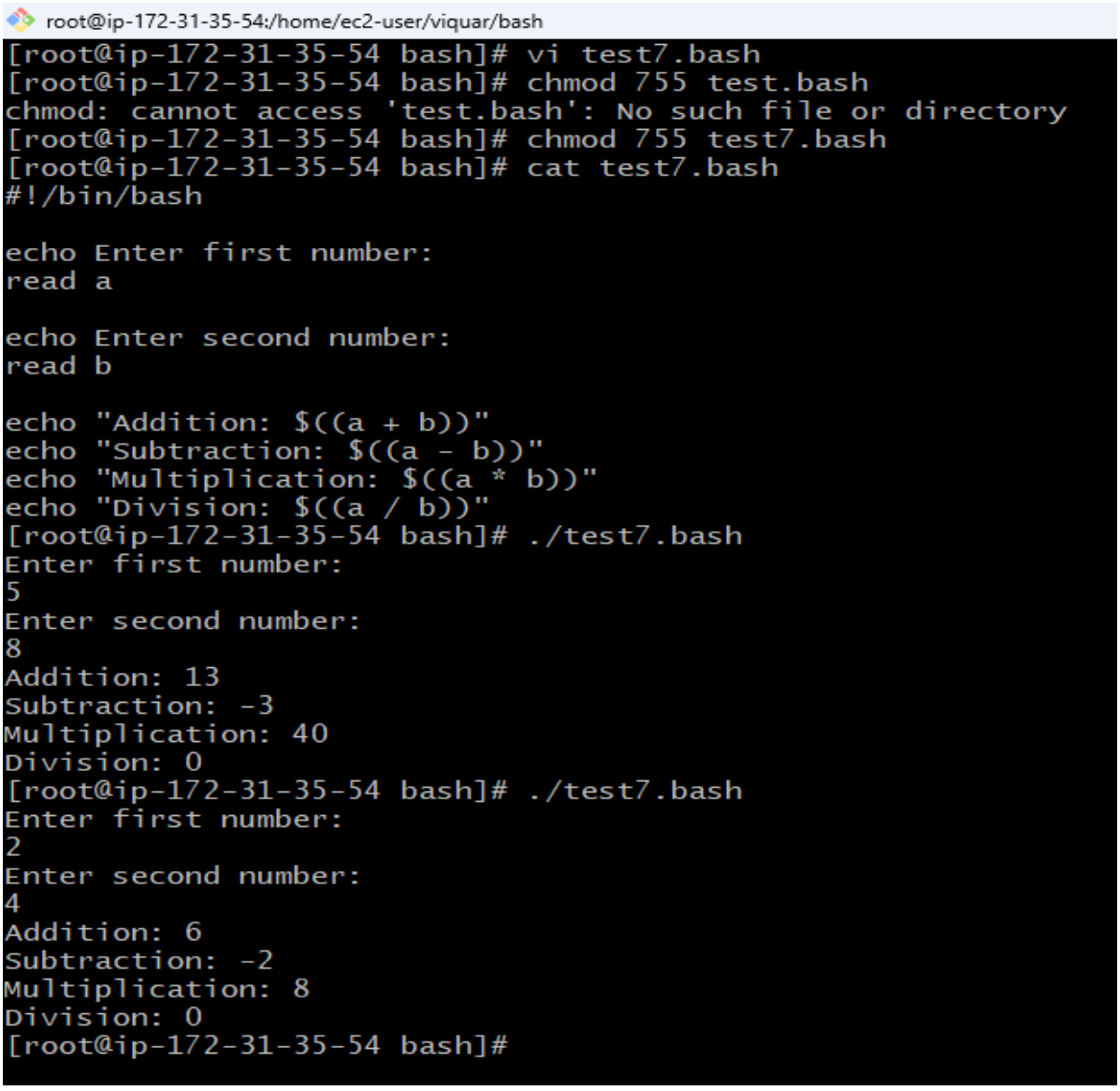
   Step 2: Change the permissions to file using a command (chmod 755 test7.bash)

   Step 3: see the contain of a file using a command (cat test7.bash)

   Step 4: Executed the script using a command (./test7.bash)

   <u>Enter the values to get results</u>

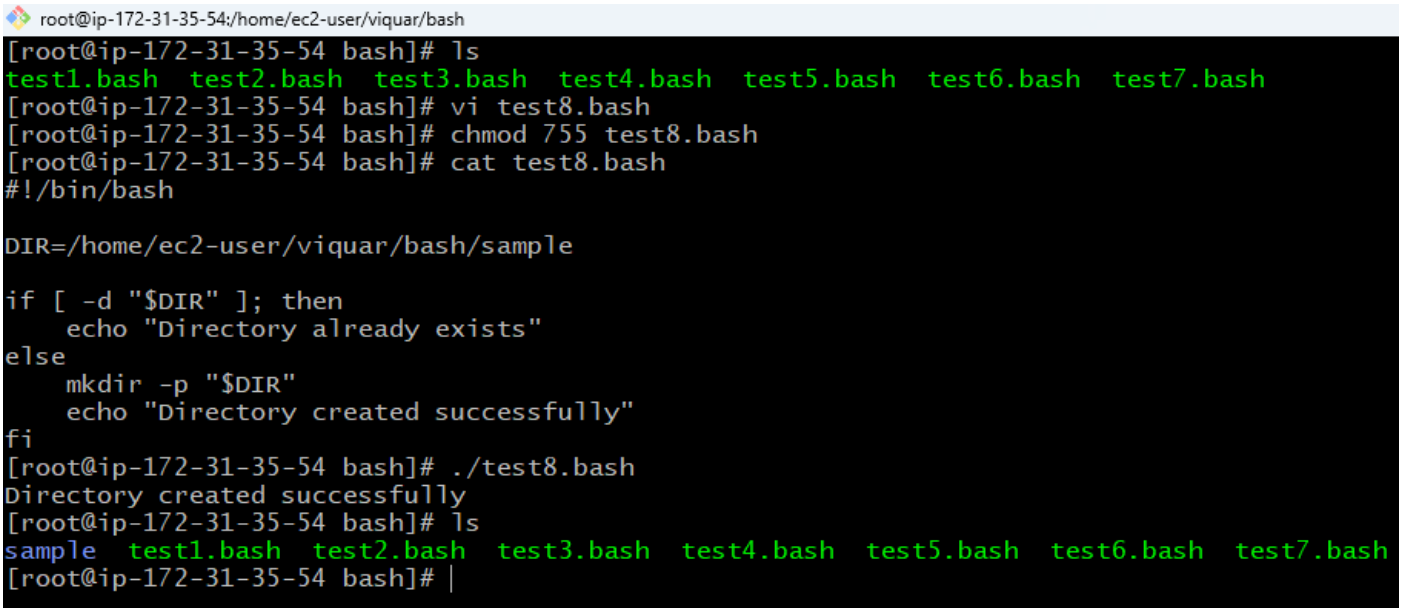
   RESULTS BELOW



8. Create a bash script to **check if a directory exists**, if not then create a directory.
   Step 1: Created a file using a command vi (test8.bash) and wrote a script
   #!/bin/bash

   DIR="/home/ec2-user/viquar/bash/sample"

   if [-d "$DIR"]; then
     echo "Directory already exists"
   else
     mkdir -p "$DIR"
     echo "Directory created successfully"
   fi
   Step 2: Change the permissions to file using a command (chmod 755 test8.bash)
   Step 3: see the contain of a file using a command (cat test8.bash)
   Step 4: Executed the script using a command (./test8.bash)
   RESULTS BELOW



9. Create a bash script to **delete the last 3 lines of a file**.
   Step 1: Created a file using a command vi (test9.bash) and wrote a script
   #!/bin/bash

   FILE=viquar.txt

   sed -i '$d' $FILE
   sed -i '$d' $FILE
   sed -i '$d' $FILE

   echo Last 3 lines deleted successfully!

Step 2: Change the permissions to file using a command (chmod 755 test8.bash)

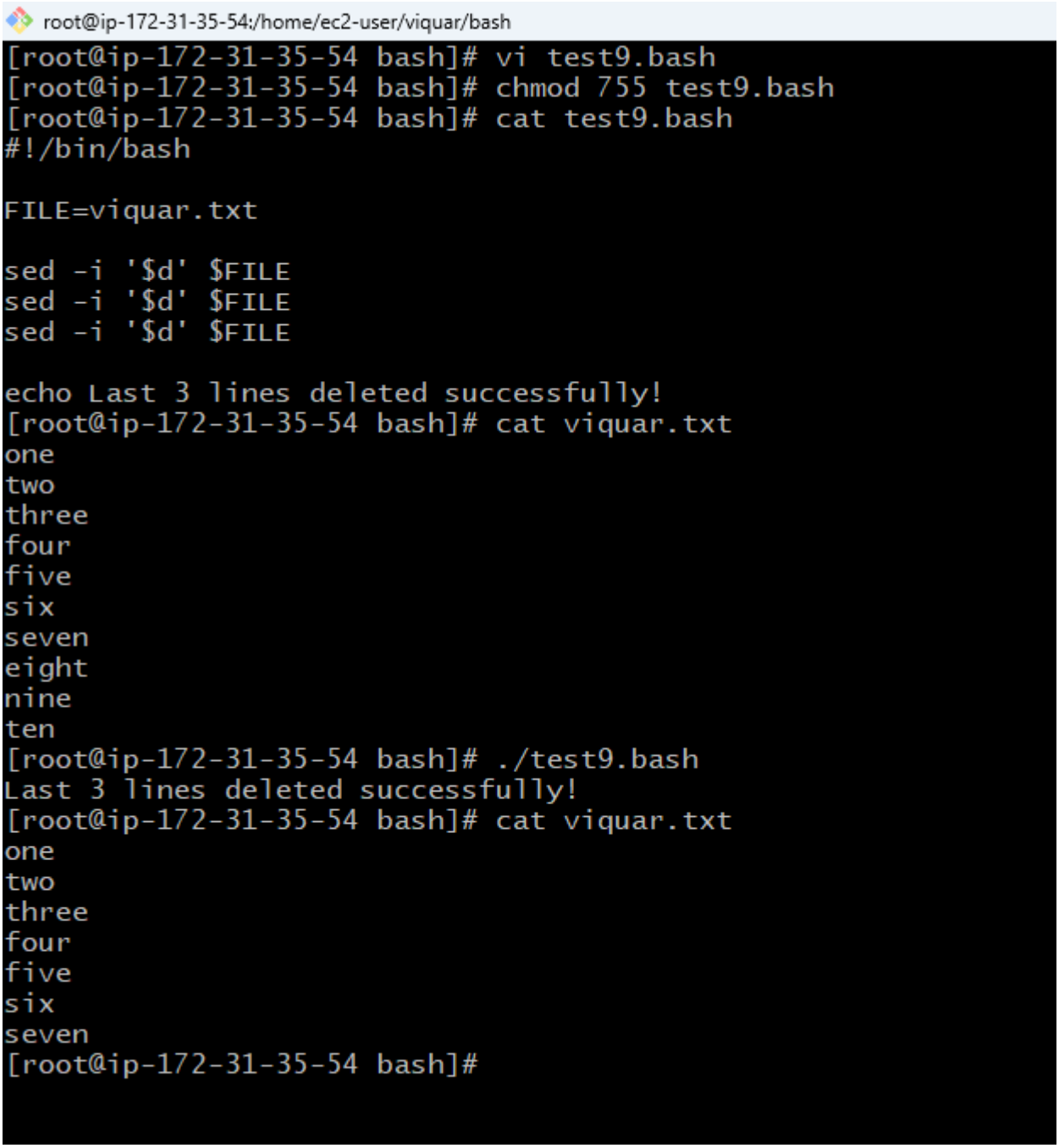
Step 3: see the contain of a file using a command (cat test8.bash)

Step 4: See the contain of viquar.txt (cat viquar.txt)

Step 5: Executed the script using a command (./test8.bash)

Step 6: See the contain of viquar.txt last 3 lines deleted or not.

RESULTS BELOW



10. Bash script to monitor cpu and if it is more than 80% then send email notification.
11. Bash script to monitor disk space and if it is more than 80% then send email notification.
12. Bash script to monitor memory and if it is more than 80% then send email notification.

Step 1: Created a file using a command vi (test10.bash) and wrote a script

```
#!/bin/bash

EMAIL="user@example.com"

# Check CPU usage
CPU=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $8}')
CPU=${CPU%.*}

if [ "$CPU" -gt 80 ]; then
   echo "CPU usage is ${CPU}%" | mail -s "CPU Alert" "$EMAIL"
fi

# Check Disk usage
DISK=$(df / | tail -1 | awk '{print $5}' | tr -d '%')

if [ "$DISK" -gt 80 ]; then
   echo "Disk usage is ${DISK}%" | mail -s "Disk Alert" "$EMAIL"
fi

# Check Memory usage
MEM=$(free | awk '/Mem:/ {printf "%.0f", $3/$2 * 100}')

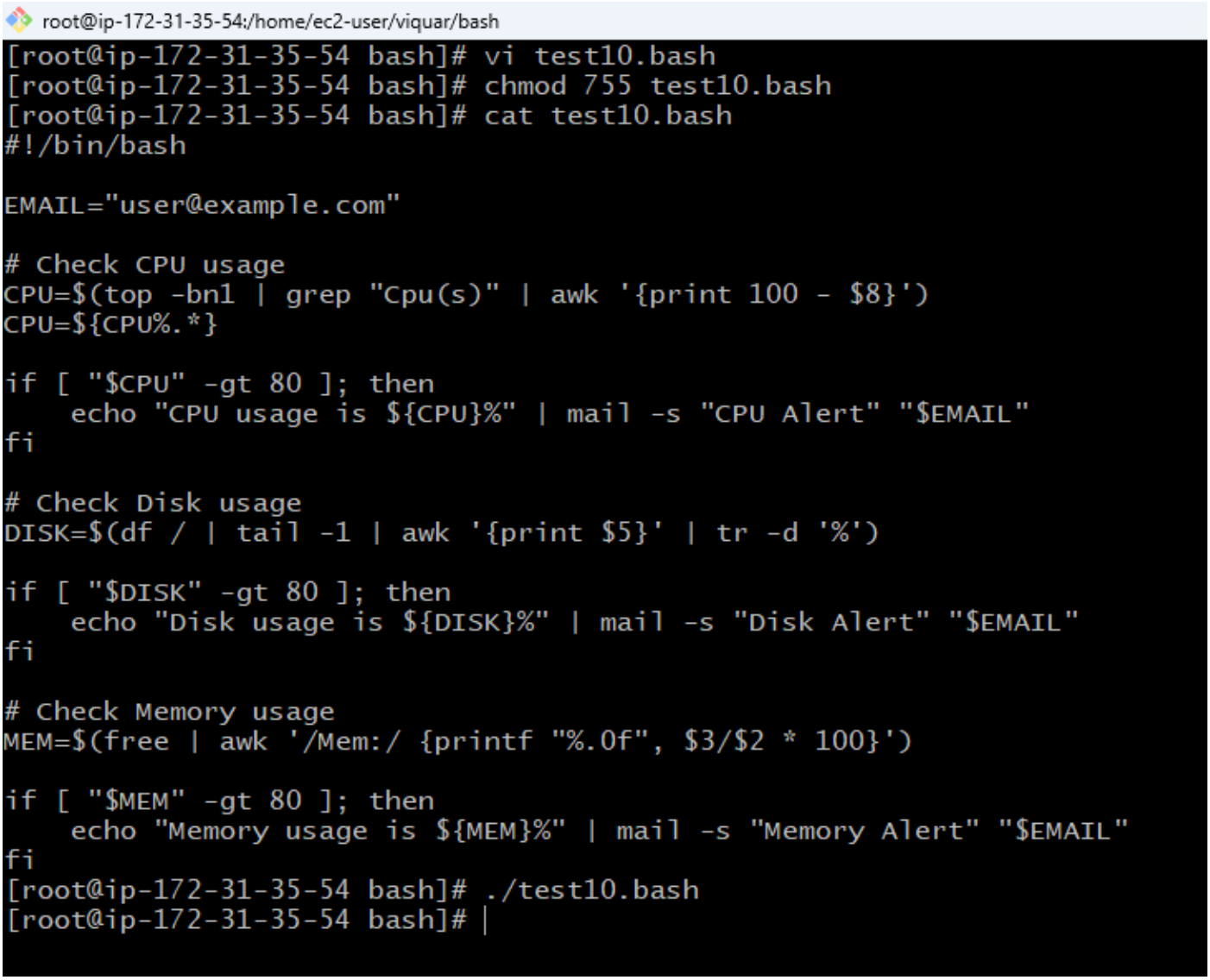
if [ "$MEM" -gt 80 ]; then
   echo "Memory usage is ${MEM}%" | mail -s "Memory Alert" "$EMAIL"
fi

echo "System monitoring completed!"
```

Step 2: Change the permissions to file using a command (chmod 755 test10.bash)

Step 3: see the contain of a file using a command (cat test10.bash)

Step 4: Executed the script using a command (./test8.bash)


root@ip-172-31-35-54:/home/ec2-user/viquar/bash
[root@ip-172-31-35-54 bash]# vi test10.bash
[root@ip-172-31-35-54 bash]# chmod 755 test10.bash
[root@ip-172-31-35-54 bash]# cat test10.bash
#!/bin/bash
EMAIL="user@example.com"
Check CPU usage
CPU=$(top -bn1 | grep "Cpu(s)" | awk '{print 100 - $8}')
CPU=${CPU%.*}
if ["$CPU" -gt 80]; then
echo "CPU usage is ${CPU}%" | mail -s "CPU Alert" "$EMAIL"
fi
Check Disk usage
DISK=$(df / | tail -1 | awk '{print $5}' | tr -d '%')
if ["$DISK" -gt 80]; then
echo "Disk usage is ${DISK}%" | mail -s "Disk Alert" "$EMAIL"
fi
Check Memory usage
MEM=$(free | awk '/Mem:/ {printf "%.0f", $3/$2 * 100}')
if ["$MEM" -gt 80]; then
echo "Memory usage is ${MEM}%" | mail -s "Memory Alert" "$EMAIL"
fi
[root@ip-172-31-35-54 bash]# ./test10.bash
[root@ip-172-31-35-54 bash]#

CRONJOBS

First check if amazon Linux have corntab installed or not

Exercise: Crontab Entries

1. April 5th Midnight
   0 0 5 4 *
2. 5th of Every November, January, June **if it is a Thursday**
   0 0 5 1,6,11, 4
3. At **05 and 27th minutes** of **9, 10, 11** hours every day
   5,27 9-11 * * *
4. **34 minutes** of **9th hour** on **15th August**
   34 9 15 8 *
5. Every midnight
   0 0 * * *
6. Every Weekend (**Saturday night 11:59 PM**)
   59 23 * * 6
7. After every reboot
   @reboot

